CHAPTER THREE

3. Theory of Consumer Behavior

Chapter objectives

After successful completion of this chapter, you will be able to:

- explain consumer preferences and utility
- differentiate between cardinal and ordinal utility approach
- define indifference curve and discuss its properties
- derive and explain the budget line
- describe the equilibrium condition of a consumer

Consumer Preferences:

What the Consumer Wants

- Our goal in this chapter is to understand how consumers make choices.

GivenThree consumption Bundles A, B & C

Bundle A: (1 Pizza, 1 Coca Cola)

Bundle B: (1 Pizza, 1 Pepsi)

Bundle C: (1 Bergur, 1 Beer)

- Which Bundle do you prefer? Rank them according to you preference.

Consumer Preferences

- Our goal in this chapter is to understand how consumers make choices.
- Consumers makes choices by comparing bundle of goods or **consumption bundles**
- **A consumption bundle** is a complete list of goods and services that are available for choice by the consumer.
- The description of **when, where and under what** circumstances the consumption bundles would become available should be known.

Consumer Preferences

Let's limit our attention to a simple choice problem

- Say our consumption bundle consists of two goods, (x_1, x_2) where x_1 is the amount of one good and x_2 is the amount of the other good
- We can denote the complete consumption bundle (x_1, x_2) by **X**
- Given another consumption bundle (y_1, y_2) denoted by **Y**

Consumer Preferences

The consumer can rank the two consumption bundles as to their desirability.

- **X $\succ$ Y ----------- Strictly preferred**
- **X$\sim$Y ----------- Indifferent**
- **X $\succeq$ Y ----------- Weakly prefers**
- If X $\succeq$ Y and Y $\succeq$ X, we can conclude that X ~Y
- If X $\succeq$ Y but we know that it is not the case that X~ Y, we can conclude that X$\succ$Y

Assumptions (axioms) of consumer preference

- These assumptions are basically made to ensure **consistency** in consumer preference. Eg. it is contradictory to say $(x_1, x_2) > (y_1, y_2)$ and at the same time $(x_1, x_2) < (y_1, y_2)$ or vice versa.

- **Complete**: any two bundles can be compared (the consumer can rank his preference or is able to make choice between any two bundles).

- **Reflexive**: we assume that any bundle is at least as good as itself; $(x_1, x_2) \geq (x_1, x_2)$.

- **Transitivity**: if $(x_1, x_2) \geq (y_1, y_2)$ and $(y_1, y_2) \geq (z_1, z_2)$, then we assume that $(x_1, x_2) \geq (z_1, z_2)$.

The concept of utility

- **Utility** is a way to describe preferences. It is a measure of the level of happiness/pleasure/satisfaction that someone receives from the consumption of a good/service.

Utility assumes that satisfaction can be measured in units, in the same way that the actual goods consumed can be determined.

- **A utility function** is a way of assigning a number to a consumption bundles such that more preferred bundles get assigned larger number than less-preferred bundles.

The concept of utility

- Given any two consumption bundles X and Y, the consumer definitely wants X than Y if and only if the utility of X is better than the utility of Y.
- **Note:**
  - *Utility' and 'Usefulness' are not synonymous*
  - *Utility is subjective*
  - *Utility can be different at different places and time*

Then, How do we measure utility?

Approaches of measuring utility

- Two major approaches to measure or compare consumer‘s utility:

1. **Cardinal Approach**
2. **Ordinal approaches.**

Approaches of measuring utility

The Cardinalist school

- First pioneered by **Alfred Marshall**, promoted by **classical and Neo-classical Economists**.

- measures utility **objectively** using utils**.** i.e. it is possible to express utility/satisfaction that an individual derives from **consuming** a commodity in quantitative terms/cardinal numbers such as 1, 2, 3, 4, 5, 6 and so on

<u>**Limitations**</u>

- **Less realistic** as quantitative measurement of utility is not possible.
- The assumption of constant **MU of money is unrealistic** because as income increases, the marginal utility of money changes.

- The assumption of cardinal utility is doubtful because utility may not be quantified. **Utility cannot be measured absolutely (objectively**).

The Ordinalist school

- was pioneered by **Prof John R Hicks,** most often propounded by **modern economists** who believe that utility being the psychological phenomenon cannot be measured theoretically, quantitatively and even cardinally.

- measures utility **subjectively.**

It is more realistic as it relies on qualitative measurement. Consumer can only **rank/order** the utility s/he derives from the consumption of different products as 1st, 2nd, 3rd, etc (satisfaction cannot be measured numerically in the form of 1, 2, 3 …).

1. The cardinal utility theory

Measurement:

- Utility is measurable by arbitrary unit of measurement (**utils.** i.e. the word crafted by Léon Walras **).** Utils help in understanding how much utility is derived from consumption of a product.
- also assumes that the utility derived from one unit of the commodity is equal to the amount of money, which a consumer is ready to pay for it (**1 Util=1 Unit of money).**
- Cardinal approach brings out the preference of one product over other through Utils but this does not imply any conclusion or relation between the choices.

1. The cardinal utility theory

Basis:

- The cardinal utility is based on **marginal utility** analysis.

Example: Donald Trump submits that Burger gives him 60 Utils of satisfaction whereas Pizza gives him only 40 Utils.

Assumptions of the cardinal utility theory

1. **Rationality** of consumers (utility maximizer - select the option that will bring the most satisfaction to them).
2. Utility is **cardinally measurable.**
3. **Constant marginal utility of money** (This assumption is critical because money is used as a standard unit of measurement of utility).
4. **Diminishing marginal utility(DMU).**
5. The total utility of a basket of goods depends on the quantities of the individual commodities.

It is assumed that consumers derive satisfaction through consumption of one good at a time.

Total and marginal utility

- **Total Utility (TU)** is the **total satisfaction** a consumer gets from consuming some specific quantities of a commodity at a particular time.
- **Marginal Utility (MU)** is the ***extra satisfaction*** a consumer realizes from an additional unit of the product.

MU = $\frac{\Delta TU}{\Delta Q}$, the slope of the TU curve

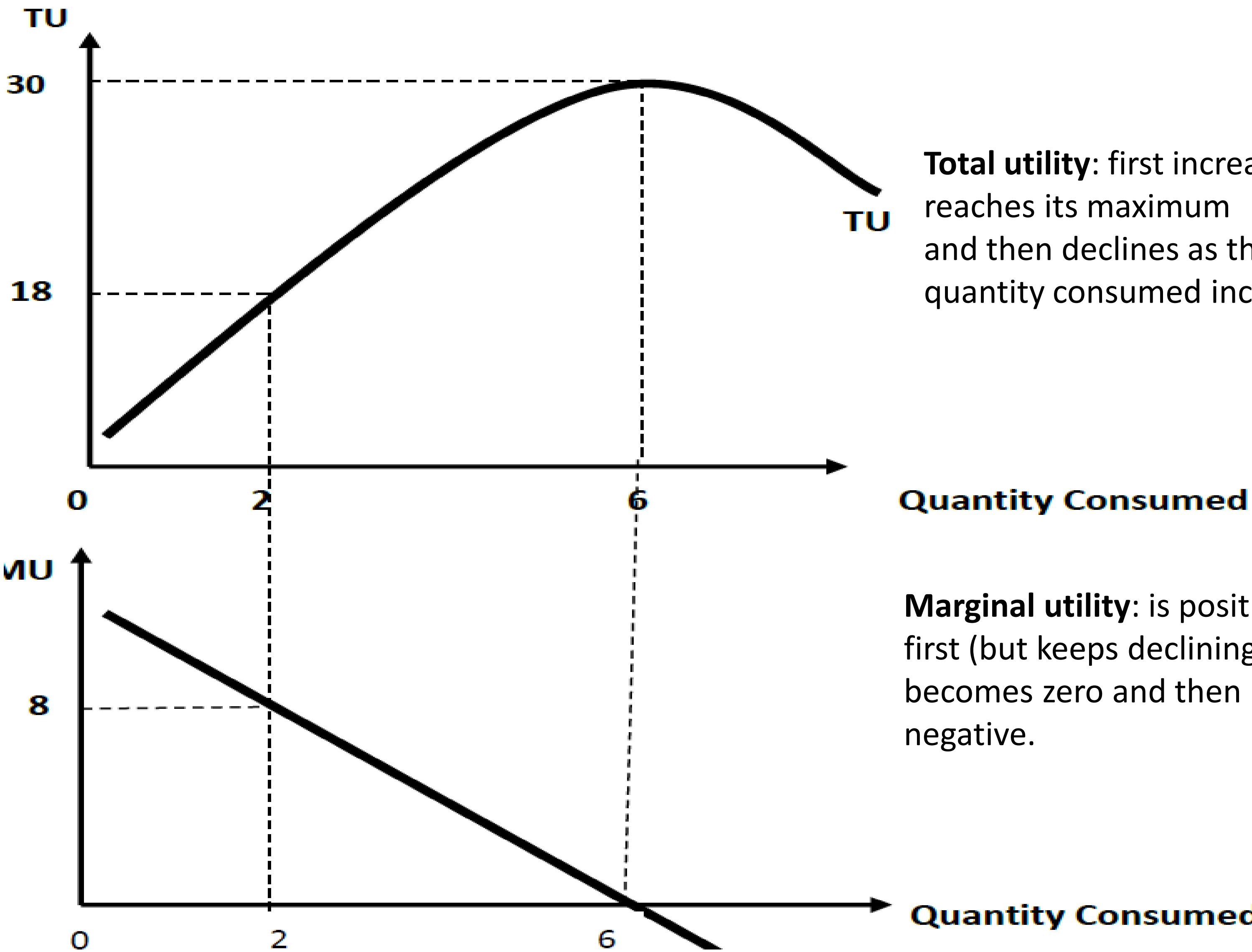
where, ΔTU is the change in total utility, and ΔQ is the change in the amount of product consumed.

Total and marginal utility

- **Consider the following**

Quantity	Total utility (TU)	Marginal utility (MU)
0	0	-
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

Total and marginal utility curves



Total utility: first increases, reaches its maximum and then declines as the quantity consumed increases.

Marginal utility: is positive at first (but keeps declining) becomes zero and then negative.

Law of diminishing marginal utility (LDMU)

- **Assuming:**
  - The consumer is rational
  - The consumer consumes identical or homogenous product (similar quality, color, design, etc.)
  - There is no time gap in consumption of the good
  - The consumer taste/preferences remain unchanged
- LDMU states that as the **quantity consumed of a commodity increases per unit of time**, the **utility derived from each successive unit decreases**.

In other words, the extra satisfaction that a consumer derives declines as s/he consumes more and more of the product in a given period of time.

Equilibrium of a consumer

- Given his **limited income** and the **price level** of goods and services, what combination of goods and services should he consume so as to get the **maximum total utility**?

a) The Case of one commodity(X):

- The consumer can either buy X or retain his income money income M
- The equilibrium condition of a consumer that consumes a single good X occurs when the marginal utility of X is equal to its market price. $MU_x = P_x$

Equilibrium of a consumer

- **If** $MU_x > P_x$, the consumer can increase his welfare by increasing by purchasing more units of x.
- **If** $MU_x < P_x$, the consumer can increase his total satisfaction by cutting down the quantity of x and keeping more of his income unspent. Therefore, he/she attains the maximization of his/her utility when $MU_x = P_x$.

Equilibrium of a consumer

b) **The Case of two commodities(X and Y):**

$$\frac{MU_x}{P_x} = \frac{MU_y}{P_y}$$

c) **The Case of n- commodities:**

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2} = \frac{MU_3}{P_3} = \cdots = \frac{MU_n}{P_n}$$

Equilibrium of a consumer

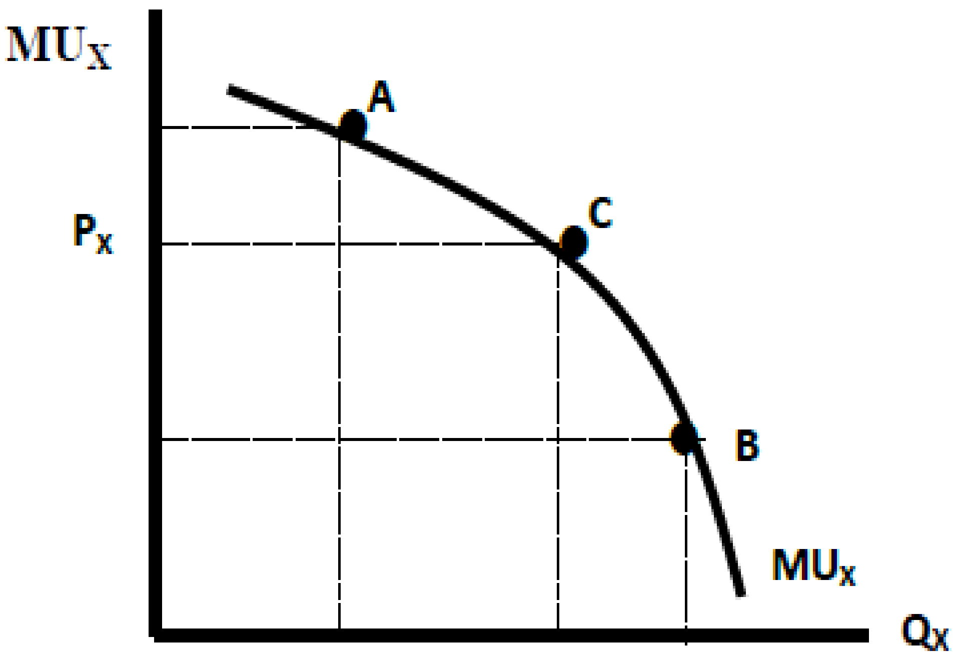


Figure 3.2: Equilibrium condition of consumer with only one commodity

Example 1

- **Example:** Suppose Saron has 7 Birr to be spent on two goods: banana and bread. The unit price of banana is 1 Birr and the unit price of a loaf of bread is 4 Birr. The total utility she obtains from consumption of each good is given below.

Income = 7 Birr, Price of banana = 1 Birr, Price of bread = 4 Birr							
Banana				*Bread*			
Quantity	*TU*	*MU*	*MU/P*	*Quantity*	*TU*	*MU*	*MU/P*
0	0	-	-	0	0	-	-
1	6	6	6	1	12	12	3
2	11	5	5	2	20	8	2
3	14	3	3	3	26	6	1.5
4	16	2	2	4	29	3	0.75
5	16	0	0	5	31	2	0.5
6	14	-2	-2	6	32	1	0.25

Example 2. Two products, x and y, are priced at $1 and $2 respectively. It is assumed that a consumer has $10 to spend.

Quantity	Product x ($1 each)				Product y ($2 each)		
	TU	MU	MU/P		TU	MU	MU/P
1	60				42		
2	100				74		
3	125				98		
4	137				106		
5	139				120		
6	141				123		

Consumer equilibrium

Two products, x and y, are priced at $1 and $2 respectively. It is assumed that a consumer has $10 to spend.

Quantity	Product x ($1 each)				Product y ($2 each)		
	TU	MU	MU/P		TU	MU	MU/P
1	60	60	60		42	42	21
2	100	40	40		74	32	16
3	125	25	25		98	24	12
4	137	12	12		106	18	9
5	139	5	5		120	14	7
6	141	2	2		123	12	6

Consumer equilibrium

Explaining the solution

- Consumer equilibrium is where the MU/P is the same for each product.
- This is where 4 units of good x and 3 units of good y are consumed with a total utility score of 235.
- No other combination of goods generates a higher total utility; therefore, any other combination would provide less satisfaction.
- This is referred to as the equi-marginal principle

Limitation of the Cardinalist approach

1. The assumption of cardinal utility is extremely doubtful.
2. The assumption of constant utility of money is also unrealistic.
3. Finally, the axiom of diminishing marginal utility has been 'established' from introspection, it is a psychological law which must be taken for granted.

2. The ordinal utility theory

Measurement:

- Utility is measured in terms of **ranking** of preferences of a commodity when compared to each other.

A consumer can tell **subjectively** whether a product derives more/less/equal satisfaction when compared to another.

Ordinal utility approach gives a sense of **preferences** (likes and dislikes) and is used in grading the preferences of consumer depending upon the alternatives that are available to him or her.

2. The ordinal utility theory

Basis:

- The concept of ordinal utility is based on **indifference curve** analysis.
- The indifference curve assumes that the utility can only be expressed **ordinally.**

Example: Donald Trump submits that he gets more satisfaction from Burger as compared to Pizza.

Assumptions of the ordinalist approach

1. The consumer is a **rational** being who aims at maximizing his level of satisfaction for given income and prices of goods and services, which he wish to consume.
2. Utility is **ordinal**.
3. **Diminishing marginal rate of substitution.**
4. The consumer is expected to make decisions that are **consistent** with his objective (Consumer's preferences are consistent/transitive)

Assumptions of the ordinalist approach

5. **Total utility** of a consumer is measured by the quantities of all items he/she consumes from his/her consumption basket.

$$U = f(x_1, x_2, x_3, \ldots, x_n)$$

– A consumer may derive satisfaction from the consumption of a combination of **goods** and **services**.

– The consumer has not reached the saturation point of any commodity and hence he prefers larger quantities of all commodities.

Indifference set

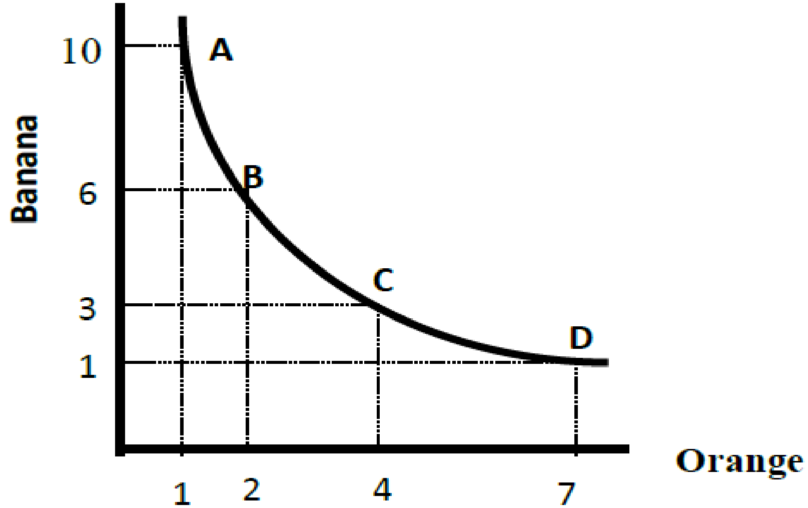
Indifference set/schedule:

- Shows a combination of goods for which the consumer is indifferent / shows the **various combinations of goods** from which the consumer **derives the same level of satisfaction**.

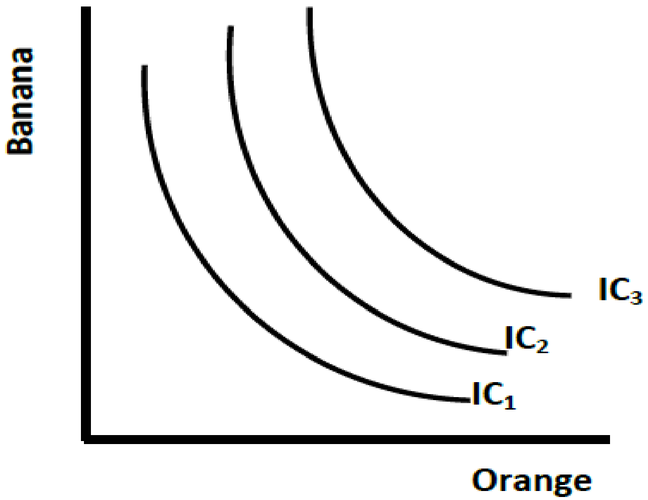
Bundle (Combination)	A	B	C	D
Orange	1	2	4	7
Banana	10	6	3	1

Indifference curve and map

- **Indifference curve:** graphically shows different combinations of two goods which yield the same utility (level of satisfaction) to the consumer.
- A set of indifference curves is called **indifference map**.



i) Indifference curve



ii) Indifference map

Properties of indifference curves

- Indifference curves have **negative slope** (downward sloping to the right).
- Indifference curves are **convex** to the origin (because of diminishing marginal rate of substitution).
- A higher indifference curve is **always** preferred to a lower one.
- Indifference curves **never cross** each other (cannot intersect).

Marginal rate of substitution (MRS)

- **Marginal rate of substitution** is a rate at which consumers are willing to substitute one commodity for another in such a way that the consumer remains on **the same level of satisfaction**.
- Marginal rate of substitution of X for Y $(MRS_{x,y})$ is defined as the number of units of commodity Y that must be given up in exchange for an extra unit of commodity X so that the consumer maintains on the same level of satisfaction (IC curve).

Marginal rate of substitution (MRS)

$$MRS_{X,Y} = \frac{Number\ of\ units\ of\ Y\ given\ up}{Number\ of\ units\ of\ X\ gained} = -\frac{\Delta Y}{\Delta X}$$

- $Slope\ of\ IC = \frac{\Delta Y}{\Delta X}$
- $MRS_{X,Y}$ is the negative of the slope of an indifference curve.

$$MRS_{X,Y} = -\frac{\Delta Y}{\Delta X}$$

- The $\boldsymbol{MRS_{X,Y}}$ is related to the $\boldsymbol{MU_X}$ and the $\boldsymbol{MU_Y}$.

MU and MRS

- . **Given** $U = f(X, Y)$, **and assume also that preferences are complete, transitive and continues.**
- Since utility is constant along an indifference curve,

$$dU = \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy = 0$$

$$\longrightarrow \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial y} dy = 0$$

$$\longrightarrow MU_x dx + MU_y dy = 0$$

MU and MRS

$$\rightarrow \quad \frac{\partial U}{\partial x} dx + \frac{\partial U}{\partial x} dy = 0$$

$$\rightarrow \quad MU_x dx = -MU_y dy$$

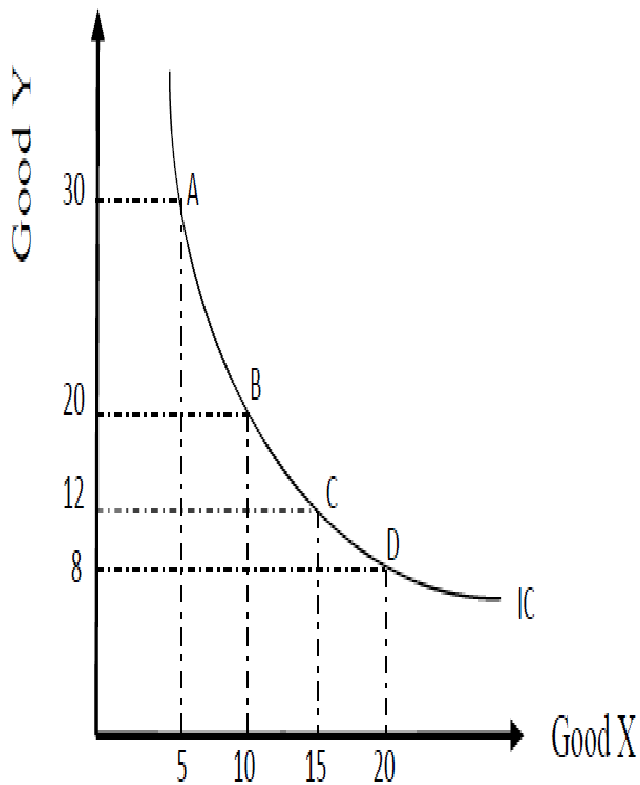
$$\rightarrow \quad \frac{MU_x}{MU_y} = -\frac{dy}{dx} = MRS_{x,y}$$ or

$$\frac{MU_y}{MU_x} = -\frac{dx}{dy} = MRS_{y,x}$$ or

so, $$\boxed{\frac{MU_x}{MU_y} = MRS_{x,y}}$$

MRS…

- **Consider the following IC:**



Calculate $MRS_{X,Y}$ as we move from point A to B, B to C and C to D

- A to B, $MRS_{X,Y} = \frac{20-30}{10-5} = -2$
- B to C, $MRS_{X,Y} = \frac{12-20}{15-10} = -1.6$
- C to D, $MRS_{X,Y} = \frac{8-12}{20-15} = -0.8$

For the same increase in consumption of good X, the amount of good Y the consumer is willing to scarify diminishes implying the **principle of diminishing marginal rate of substitution.**

MRS…

- **Example 1:** Suppose a consumer's utility function is given by $\mathrm{U}\ (\mathrm{X}, \mathrm{Y}) = X^4Y^2$. Find $MRS_{X,Y}$

Solution: $MRS_{X,Y} = \dfrac{MU_X}{MU_Y}$

$$MU_X = \frac{\partial U}{\partial X} = 4X^3Y^2 \quad and\ MU_Y = \frac{\partial U}{\partial Y} = 2X^4Y \quad \text{Hence, } MRS_{X,Y} = \frac{MU_X}{MU_Y} = \frac{4X^3Y^2}{2X^4Y} = \frac{2Y}{X}$$

- **Example 2:** Suppose a consumer's utility function is given by $\mathrm{U}\ (\mathrm{X}, \mathrm{Y}) = X^5Y^3$. Find $MRS_{Y,X}$
- **Example 3:** Given $\mathrm{U}\ (\mathrm{X}, \mathrm{Y}) = X^aY^b$, **find** $MRS_{X,Y}$**.**

Budget Constraint

- Indifference curves only tell us about the consumer's preferences for any two goods but they can not tell us which combinations of the two goods will be chosen or bought.
- In reality, the consumer is **constrained** by his/her **money income** and **prices** of the two commodities.
- In other words, individual choices are also affected by budget constraints that limit people's ability to consume in light of prices they must pay for various goods and services.

Budget Constraint

- To develop the budget constraint /budget set of the consumer we need assumptions:
  - There are only two goods bought in quantities, say, X and Y.
  - Each consumer is confronted with market determined prices, $\boldsymbol{P_X}$ and $\boldsymbol{P_Y}$.
  - The consumer has a known and fixed money **income (M).**

- Thus the budget constraint of the consumer is given by:

$$\boldsymbol{P_X X + P_Y Y \leq M}$$

Budget set and line

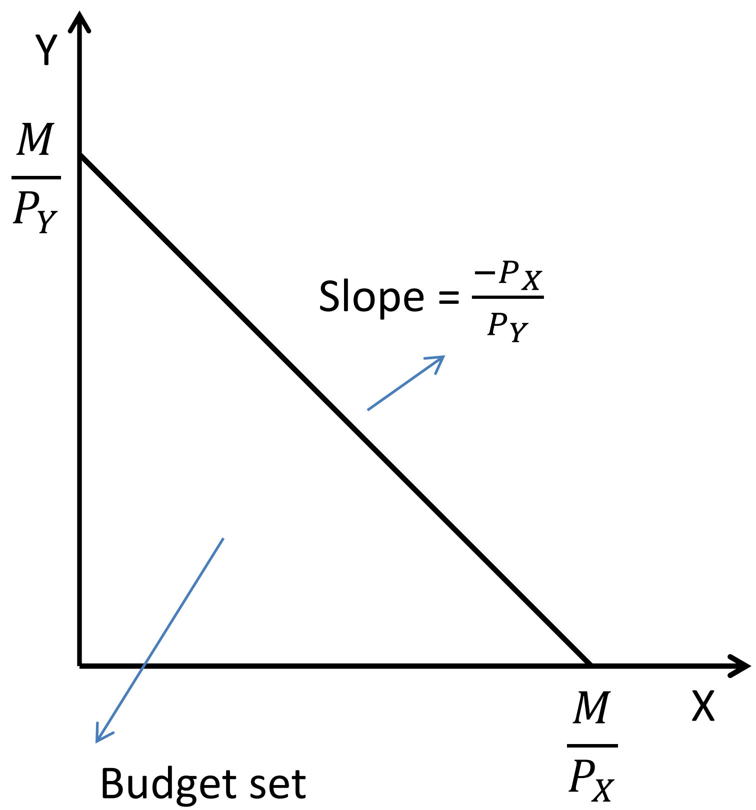
- **Budget set:** the set of affordable bundles given prices (P_x and P_y) and income M.
- **Budget line:** the set of bundles that cost exactly M.

  i.e. $\boldsymbol{P_X X + P_Y Y = M}$

To draw a budget line we have to consider the two extreme income and expenditure relationship,

i.e. **identify vertical and horizontal intercept as well as slope.**

Budget line



- Any bundle on or within the budget line is affordable.
- Any bundle outside the budget line is unaffordable.
- The budget line has negative slope because in order to consume more of good X, we have to consume less of good Y and vice versa to satisfy the budget constraint.

Examples

Example 1: A consumer has 100 birr to spend on two goods X and Y with prices 3 birr and 5 birr respectively. Derive the equation of the budget line and sketch the graph.

<u>Solution</u>

Budget equation: $3X + 5Y = 100$

Intercepts: *when* $Y=0$, $X=M/P1= 100/40= 33.4$

when $X=0$, $Y=M/P2= 100/5=20$

Slope: $-P_1/P_2 = -3/5$

Example 2: Let $M= 200$, $p_1=2$, $p_2=4$, Derive the budget equation and draw a budget line.

Determinants of the budget line

- The budget line changes if one of the following things changes;

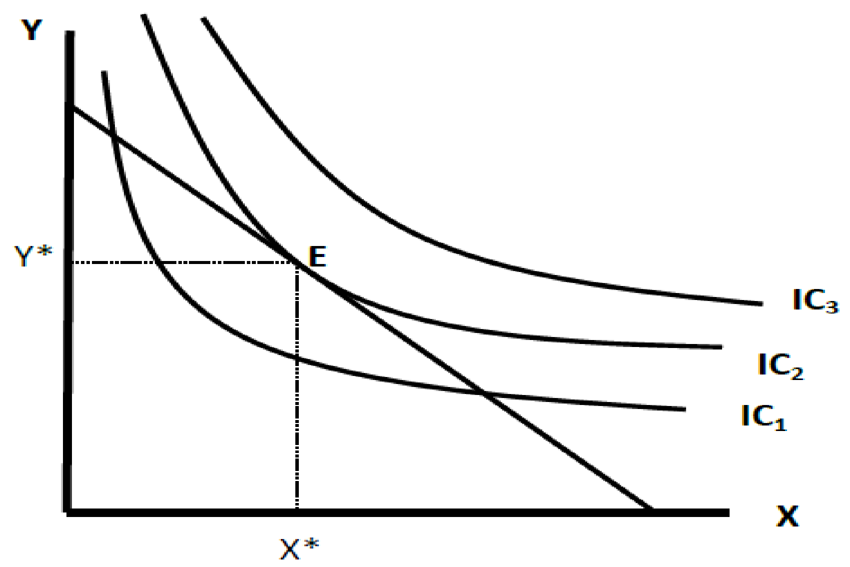
1. The consumers income
2. The prices of goods
3. Taxes, subsides and rationing

Equilibrium of the consumer
(Optimal choice) - Ordinal approach

- A rational consumer tries to attain the **highest possible indifference curve**, given the budget line. This occurs at the point where the **indifference curve is tangent to the budget line** (point E)**.**

i.e. The slope of the indifference curve ($MRS_{X,Y}$) is equal to the slope of the budget line (P_X/P_Y).

$$MRS_{X,Y} = MU_X/MU_Y = P_X/P_Y$$



Optimal choice - Ordinal approach

Example 1: A consumer consuming two commodities, X and Y, has the utility function $\mathrm{U}\ (\mathrm{X}, \mathrm{Y}) = \mathrm{XY} + 2\mathrm{X}$. The prices of the two commodities are 4 birr and 2 birr respectively. The consumer has a total income of 60 birr to be spent on the two goods.

a) Find the utility maximizing quantities of good X and Y.

b) Find $MRS_{X,Y}$ at equilibrium.

Example 2: Given a utility function $\mathrm{U}\ (\mathrm{X}, \mathrm{Y}) = \mathrm{XY}$, and the price of the two goods 2 birr and 4 birr respectively, and income of the consumer 300 birr, find the quantities of X and Y that should be consumed to maximize utility.